ITCS 6169/8169

Computer Vision

Assignment 2: Self-Supervised Visual Representation Learning

1 Objective

In Assignment 1, you explored supervised convolutional neural networks for image classification under a limited-data setting. In this assignment, we will ask a different question:

Can we learn better visual representations from the same limited dataset without relying solely on class labels?

The objective of this assignment is to achieve the highest possible image classification accuracy by using **self-supervised learning (SSL)** and other advanced representation-learning techniques discussed in class. You will work with the same small-scale image classification setting: only **2,400 training images across 16 classes**. The key challenge is to learn useful representations despite the limited amount of labeled data. **You may not use models pretrained on external datasets.** All models used for the final classification system must be trained using only the data provided for this assignment.

2 Problem Statement

You are given a dataset containing 2,400 training images belonging to 16 classes. Your goal is to obtain the highest possible classification accuracy using advanced neural-network training strategies. Unlike Assignment 1, simply training a supervised classifier is **not sufficient**. Your solution must investigate **self-supervised representation learning**. Examples include, but are not limited to:

- Contrastive learning (e.g., SimCLR, MoCo)
- Masked image modeling (e.g., MAE)
- Self-distillation (e.g., DINO)
- Joint supervised + self-supervised learning
- Pretraining with an SSL objective followed by supervised fine-tuning
- Other self-supervised objectives that you can justify

You are encouraged to experiment with different architectures, augmentations, objectives, and training schedules.

3 Important Rule: No External Pretraining

The purpose of this assignment is to understand whether useful visual representations can be learned from the available data itself. Therefore:

- **Allowed:** Randomly initialized CNNs, Vision Transformers, or other architectures.
- **Allowed:** Self-supervised pretraining using the provided training images.

- **Allowed:** Joint optimization of classification and self-supervised objectives.
- **Allowed:** Standard data augmentation and regularization techniques.
- **Not allowed:** ImageNet-pretrained CNNs or Vision Transformers.
- **Not allowed:** CLIP, DINO, DINOv2, MAE, SAM, or other foundation/pretrained models *with externally pretrained weights*.
- **Not allowed:** Additional images or external datasets for training.

You may implement an architecture inspired by a pretrained/foundation model, but its weights must be initialized from scratch and trained only on the provided data.

4 Required Experiments

At minimum, your report should compare the following:

1. **Supervised Baseline:** Train a model using only the classification objective.
2. **Self-Supervised Learning:** Use an SSL method to learn visual representations from the training images, followed by supervised classification/fine-tuning.
3. **Your Best Training Recipe:** Develop your strongest model using insights from your experiments. This may involve SSL pretraining, joint objectives, stronger augmentations, regularization, architectural changes, or other techniques.

The architecture does not necessarily need to be identical across all experiments, but your comparisons should be sufficiently controlled to support the conclusions you draw.

5 Think Like a Researcher

This assignment is not simply about calling an existing SSL implementation. You should investigate **why** a particular training strategy works or does not work in this limited-data regime. For example:

- Does SSL actually improve classification accuracy with only 2,400 images?
- Which augmentations are useful for learning representations on this dataset?
- Does SSL pretraining followed by fine-tuning outperform joint supervised + SSL training?
- Does the choice of backbone affect the benefit of SSL?
- Does SSL reduce overfitting?
- Is a more complicated SSL method necessarily better?

You do not need to investigate all of these questions. Choose experiments that help you understand and improve your system.

6 Contents of the Report

Submit a report with a maximum length of **two pages**. The report should be concise and focus on your most important experiments and observations. The report must include the following:

- **Best Accuracy and GitHub Repository:** Clearly report the highest test-set accuracy achieved by your system. Include a link to your GitHub repository containing the code, trained model/checkpoint, inference code, and instructions required to reproduce your result.

- **Supervised Baseline:** Describe your supervised baseline, including the architecture and important training choices. Report its classification accuracy.
- **Self-Supervised Learning Strategy:** Explain the SSL objective you used and how representations were learned. Clearly describe whether you used SSL pretraining followed by fine-tuning, joint optimization, or another strategy.

  Do not simply state that you "used MoCo," "used DINO," or "used MAE." Explain the important idea behind the objective and how you adapted it to this dataset.
- **Experimental Analysis:** Compare your supervised baseline, SSL approach, and final model. Include a compact table summarizing your most important experiments.

  For example:

Method	**Test Accuracy**
Supervised Baseline	XX.X%
SSL + Fine-tuning	XX.X%
Joint SSL + Classification	XX.X%
Best Model	XX.X%

  More important than the number of experiments is your interpretation of the results. Explain **what worked, what did not work, and why you think this happened**.
- **Limitations and Conclusion:** Discuss the major limitations of your approach and summarize your key observations. What did you learn about representation learning when labeled data is limited?

7 Use of AI Coding Tools

The use of AI coding tools is **allowed and encouraged** for this assignment. You may use tools such as ChatGPT, GitHub Copilot, Claude, Gemini, or other coding assistants to help:

- implement models and training pipelines,
- debug code,
- understand unfamiliar APIs,
- implement ideas from research papers,
- analyze experimental results, and
- improve code organization and documentation.

However, **AI is your pair programmer, not your research scientist.** You are responsible for deciding:

- which SSL method to investigate,
- which experiments to perform,
- how to design fair comparisons,
- how to interpret the results, and
- what conclusions can be supported by your experiments.

You should understand all major components of your submitted code and be able to explain your design choices and results.

8 GitHub and Reproducibility

Your GitHub repository should contain:

- training code,
- inference/evaluation code,
- model definitions,
- trained checkpoint for your best model,
- configuration files or hyperparameters,
- a `README.md` with clear reproduction instructions.

A grader should be able to understand how your model was trained and reproduce the reported evaluation from the repository. Large checkpoints may be hosted externally using services such as Google Drive or Hugging Face, with the appropriate link included in the repository.

9 Grading Rubric

The assignment will be evaluated according to the following criteria:

Component	**Weight**
Classification Performance	20%
SSL / Representation Learning Approach	30%
Experimental Design and Analysis	25%
Technical Creativity and Intuition	15%
Reproducibility and Report Quality	10%

Accuracy matters, but **the highest accuracy alone does not guarantee the highest grade**. We are particularly interested in whether your experiments demonstrate thoughtful use and understanding of self-supervised representation learning. A simple but well-motivated approach with careful experiments may receive a higher grade than a complicated pipeline without meaningful analysis.

10 Computational Resources

PyTorch is recommended for implementing your solution. You may use Google Colab or the educational computing resources available for the course. If you require additional computational resources, contact the instructional staff. You are responsible for complying with the policies of any external computing resources that you choose to use.

11 Competition and Collaboration

Friendly competition is encouraged. You are welcome to discuss high-level ideas and research papers with classmates, but your implementation, experiments, analysis, and report must be your own. Do not share trained models, code, hyperparameters, or detailed experimental results with other students before the submission deadline.

12 Submission

Submit a **maximum two-page PDF report through Canvas**. The report must contain a working link to your GitHub repository. You do not need to separately upload your source code to Canvas.

Deadline: October 30, 2026 at 11:59 PM EDT

Late submissions will follow the course late-submission policy.

Good luck! Learn useful representations, not just labels.